NLP Part 4: Normalization

What is Normalization?

Normalization is the process of converting text into a **standard and consistent form**.

Example:

HELLO! → hello

Common normalization techniques include:

- Lowercasing
- Stemming
- Lemmatization

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1. Stemming

Stemming removes prefixes or suffixes from a word to produce a **root-like form**.

Examples:

Original Word	Stem
playing	play
played	play
studies	studi
connected	connect

Important: The stem may **not be a real dictionary word**.

Example:

studies → studi

---

2. Lemmatization

Lemmatization converts a word into its **correct dictionary base form**, called a **lemma**.

Examples:

Original Word	Lemma
playing	play
studies	study
running	run
better	good

Stemming vs. Lemmatization

Stemming	Lemmatization
Cuts word endings	Finds the correct base word
Faster	Usually slower
May produce invalid words	Produces meaningful dictionary words
Less linguistically accurate	More linguistically accurate
studies → studi	studies → study

Which One is Better?

Lemmatization is generally better when we want **accurate and meaningful words**.

Stemming can be useful when we want **simple and fast text processing**.

Easy way to remember:

> **Stemming cuts the word; Lemmatization understands the word.**

In [10]:
```
import string

text = "Hello! I LOVE Natural Language Processing."

# Convert to lowercase
text = text.lower()

# Remove punctuation
text = text.translate(str.maketrans('', '', string.punctuation))

print(text)
```

```
hello i love natural language processing
```

Stemming 2

In [11]:
```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

words = [
    "playing",
    "played",
    "plays",
    "studies",
```

```
    "studying",
    "connected",
    "connection"
]

for word in words:
    print(word, "->", stemmer.stem(word))
```

```
playing -> play
played -> play
plays -> play
studies -> studi
studying -> studi
connected -> connect
connection -> connect
```

In [12]:
```
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize

stemmer = PorterStemmer()

text = "The students are studying and playing with connected computers."

tokens = word_tokenize(text)

for word in tokens:
    print(word, "->", stemmer.stem(word))
```

```
The -> the
students -> student
are -> are
studying -> studi
and -> and
playing -> play
with -> with
connected -> connect
computers -> comput
. -> .
```

Lemmatization (3)

In [13]:
```
from nltk.stem import WordNetLemmatizer

#nltk.download('wordnet')
#nltk.download('omw-1.4')

lemmatizer = WordNetLemmatizer()

words = [
    "playing",
    "played",
    "studying",
    "studies",
    "running"
]

for word in words:
    lemma = lemmatizer.lemmatize(word, pos='v')
    print(word, "->", lemma)
```

```
playing -> play
played -> play
studying -> study
studies -> study
running -> run
```

In [14]:
```
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

words = ["studies", "studying", "playing", "connections"]

print("WORD\t\tSTEM\t\tLEMMA")
print("-" * 40)

for word in words:
    stem = stemmer.stem(word)
    lemma = lemmatizer.lemmatize(word, pos='v')

    print(word, "\t", stem, "\t\t", lemma)
```

```
WORD            STEM            LEMMA
----------------------------------------
studies          studi           study
studying         studi           study
playing          play            play
connections      connect                 connections
```